

| V <sub>ref</sub> | V <sub>IN</sub> | V <sub>OUT</sub> | ΔV <sub>OUT</sub> | I <sub>MAX</sub> | ΔI <sub>OUT</sub> | f <sub>sw</sub> | η   | D     | datasheet            |
|------------------|-----------------|------------------|-------------------|------------------|-------------------|-----------------|-----|-------|----------------------|
| 0.8 V            | 24 V            | 14.5 V           | 0.725V            | 3 A              | 1.5A              | 500 kHz         | 95% | 0.604 | <a href="#">link</a> |

Assuming: R10 = 10.2k

$$R9 = 174675 \approx 174k \quad 8.2.2.2 \quad \frac{V_{out}-V_{ref}}{V_{ref}} \times R10$$

$$C14 = 100nF \quad 8.2.2.3$$

$$K = 40\% \quad 8.2.2.5 \quad \frac{\Delta I_L}{I_{OUT\_MAX}}$$

$$L1 = 9.57\mu H \approx 10\mu H \rightarrow 15\mu H \quad 8.2.2.5 \quad \frac{V_{IN} - V_{OUT}}{f_{SW} \times K \times I_{MAX}} \times \frac{V_{OUT}}{V_{IN}}$$

But because in 8.2.2.6, Table 8-2 says for 12V output 12uH inductor is recommended I chose 15uH for my 14.5V output (webench has 10uH)

$$\Delta I_L = 1.2 \text{ A} \quad V_{IN\_MAX} = V_{IN} \quad 8.2.2.5 \quad \frac{V_{IN\_MAX} - V_{OUT}}{L \times f_{SW}} \times \frac{V_{OUT}}{V_{IN\_MAX}}$$

$$I_{L\_PEAK} = 3.6A \quad 8.2.2.5 \quad I_{OUT} + \frac{\Delta I_L}{2}$$

$$I_{L\_RMS} = 3.02A \quad 8.2.2.5 \quad \sqrt{I_{MAX}^2 + \frac{\Delta I_L^2}{12}}$$

$$C_{OUT} \geq 5.93\mu F \quad 8.2.2.6 \quad \frac{\Delta I_{OUT}}{f_{SW} \times \Delta V_{OUT} \times K} \times \left[ (1 - D) \times (1 + K) + \frac{K^2}{12} (2 - D) \right]$$

C<sub>OUT</sub> are C41 and C42 with 32uF in total

Input caps are from webench

Max current, assuming R<sub>θJA</sub> = 40 C/W (Junction- Board is 20 + 20 as copper plane to ambient) T<sub>A</sub> = 25 and T<sub>J</sub> = 100

$$I_{OUT\_MAX} = 2.46 \text{ A} \quad 8.2.2.9 \quad \frac{(T_J - T_A)}{R_{\theta JA}} \times \frac{\eta}{1 - \eta} \times \frac{1}{V_{OUT}}$$

| $V_{IN}$ | $V_{OUT}$ | $I_{MAX}$ | $f_{sw}$ | $\eta$ | datasheet            |
|----------|-----------|-----------|----------|--------|----------------------|
| 14.5 V   | 7 V       | 2 A       | 650kHz   | 95%    | <a href="#">link</a> |

Assuming:  $R_{13} = 10.2k$

$$R_{12} = 83133 \approx 84.5k \quad 8.2.1.2.1 \quad \left(\frac{V_{OUT}}{0.765} - 1\right) \times R_{13}$$

$L_2 = 6.8 \mu H$  according to Table 2 in 8.2.1.2.2

I chose greater than  $4.7 \mu H$  to get lower current fluctuations

8.2.1.2.2

$$I_{P-P} = 0,859 A \quad \frac{V_{OUT}(V_{IN(MAX)} - V_{OUT})}{V_{IN(MAX)} \times L_O \times f_{SW}}$$

$$I_{PEAK} = 2.43 A \quad I_{MAX} + \frac{I_{P-P}}{2}$$

$$I_{IO(RMS)} = 2.02 A \quad \sqrt{I_{MAX}^2 + \frac{I_{P-P}^2}{12}}$$

$$I_{CO(RMS)} = 0.236 A \quad \frac{V_{OUT} \times (V_{IN} - V_{OUT})}{\sqrt{12} \times V_{IN} \times L_O \times f_{SW}}$$

Rest is from datasheet

Input cap 8.2.1.2.3

Bootstrap cap 8.2.1.2.4